

Xing Zhou

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Education

Beijing University of Posts and Telecommunications · Undergraduate — Sep 2024 – Jun 2028

School of Computer Science (National Pilot Software Engineering School) · Software Engineering · Year 3

Major rank **5/156** GPA **3.78/4.0** Weighted avg. **90.96/100** CET-6 **644/710**

Selected coursework (out of 100): Discrete Mathematics 98; Probability and Mathematical Statistics 97; C++ Programming 98; Computer Organization and Architecture 95; Operating Systems 92.

Research Experience & Academic Training

Peng Xu's Research Group — Research Trainee — Oct 2025 – Jan 2026

Explored how AI agents organize, store and retrieve memories through literature review, analysis of memory-management mechanisms and development of a demonstration prototype.

- **Literature review:** Reviewed *Generative Agents* and related studies, examining memory streams, planning and reflection, and hierarchical retrieval; summarized findings in a research presentation.
- **Prototype:** Analyzed retrieval scoring based on recency, importance and relevance; developed a demonstration prototype for memory management using Flask, MySQL and Vue.

Selected Projects

ClusterFlow ↗ — Team & Technical Lead — Sep 2026 – Present

A digital twin platform for simulating and validating heterogeneous AI compute scheduling, covering resource modeling, task DAGs, scheduling simulation, run records and observability.

- **Supervisor:** Implemented the Rust/Linux Supervisor, including UDS session handling, permission checks, subprocess lifecycle management and safe resource cleanup.
- **Integration:** Led platform module decomposition and cross-language interface design; coordinated the integration of scheduling and simulation components.

RepoPilot ↗ — Sole Developer — Jun 2026 – Jul 2026

A repository reconnaissance tool with CLI and MCP entry points, enabling AI agents to inspect repository structure, read project documentation and manage analysis reports.

- **Tooling:** Built Python/FastMCP tool interfaces and routing for controlled repository inspection and analysis.
- **Safety:** Implemented read-only permission boundaries, credential isolation and path validation; added cleanup dry runs with explicit confirmation. Released two versions.

SpiritInk ↗ — Team Lead & Core Developer — Mar 2026 – Apr 2026

Uses OpenCV to process visual hand input, combining stroke trajectories and character recognition for touch-free virtual calligraphy with real-time interaction feedback.

- **Interaction:** Built the C++ main loop, stroke handling, coordinate mapping, character assembly and undo.
- **Integration:** Linked the C++ runtime to Electron/React; fixed shutdown issues, stalls and feedback bugs.

Open-Source Contributions

Microsoft DocumentDB for VS Code ↗ — 3 merged PRs — May 2026

#670 MongoDB shard-key display **#674** Hidden-index tree and menus **#679** Safer deletion confirmation

Honors & Awards

First-Class Undergraduate Merit Scholarship; Merit Student (both university-level) — AY 2024–2025

Second Prize, National English Competition for College Students, Category C

Digital China Corporate Scholarship; Merit Student (both university-level) — AY 2025–2026

Second Prize, 3rd BUPT Hongyan Cup Student Innovation Competition (Chuyan Track)

Technical Skills

Languages: C/C++, Python, Rust, Java, C#, Kotlin, JavaScript/TypeScript, SQL, Shell/Bash.
Development: Linux/WSL2, Git, MySQL/SQLite, Spring Boot, Flask/FastAPI, Vue/React, Electron.
AI & Engineering: OpenCV, MCP/FastMCP, LLM agent tooling, systems integration and automated testing.